

RIDHI BANDARU

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she/they

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Phone: +1 (636) 856-7436

Education

University of Connecticut, Storrs

M.S., Data Science, 2024

GPA: 3.93/4.00

Birla Institute of Technology and Science, Pilani – Hyderabad Campus

B.E., Electronics and Instrumentation Engineering (Honors), 2023

Minor, Robotics and Automation

CGPA: 8.27/10.00

Research Experience

Lab Manager / Predoctoral Researcher

w/ Prof. Alexandra Decker

June 2026 – Present

*How We Learn Lab, Washington University in St. Louis***Independent Research Collaborator**

w/ Dr. Abhishek Dedhe & Prof. Jessica Cantlon

February 2025 – Present

Kid Neuro Lab, Carnegie Mellon University

- Designed and implemented a Language-of-Thought framework with a cognitively grounded Domain-Specific Language to model hierarchical sequence-generation strategies

Peer-reviewed Publications

Choudhary, M., Agrawal, A.*, **Bandaru, R.***, Dedhe, A.*, Choudhury, M. (Accepted as a full paper). **Stochastic Parrot See, Stochastic Parrot Do: Hierarchical Sequence Processing Across Artificial and Biological Intelligences.** Proceedings of the Annual Meeting of the Cognitive Science Society 48.

Works in Progress

Bandaru, R., Dedhe, A., Ferrigno, S.M., Piantadosi, S.T., Cantlon, J.F. *In prep.* **One Cognitively Plausible Language of Thought for Hierarchical Sequence Generation**

Choudhary, M., **Bandaru, R.***, Agrawal, A.*, Choudhury, M., Dedhe, A. *In prep.* **Parallels in Hierarchical Composition: Human and Machine Sequence Processing Compared**

Presentations

Refereed talks

Bandaru, R., Dedhe, A. **Estimating the Price of Hierarchical Reasoning.** June 2, 2026. Diverse Intelligences Summit 2026. (In-person, Princeton University)

Lab presentations

Bandaru, R. **LoT-Modeling Sequential Generation Tasks.** October 23, 2025. Concepts, Actions, and Objects Lab, Carnegie Mellon University. (Virtual)

Posters

Bandaru, R., Dedhe, A. **The price of hierarchical reasoning.** June 2, 2026. Canadian Society for Brain, Behaviour and Cognitive Science – Annual Meeting 2026. (Accepted, did not present)

Scholarships

Independent Study Grant of \$1325 (Fall 2023). Department of Chemistry, University of Connecticut

Annual Departmental Scholarship of \$7855 (Fall 2023). Department of Applied Mathematics and Statistics, Johns Hopkins University (did not accept)

Undergraduate Teaching Award of ₹20000 (Spring 2023). Academic-Undergraduate Studies Division, BITS Pilani – Hyderabad Campus

Mentorship

- Helen Fang (Troy High School, CA → University of California, Berkeley)
- Brandon Lee (Troy High School, CA)
- Michael Xiong (Troy High School, CA → Carnegie Mellon University)
- Samantha Juteram (University of California, Irvine / Centene Corporation)

Teaching

Tutor (Spring 2024), **Undergraduate Math & Statistics, Q-Center, University of Connecticut**

- Provided one-on-one and group tutoring sessions to undergraduate students in graduate-level courses
- Developed weekly mini-lessons and offered structured support for assignments and exam preparation

Teaching Assistant (Spring 2023), **CS F327 Computer Architecture, BITS Pilani, Hyderabad Campus**

- Lab, 40 students
- Assisted students with Verilog simulations of RISC-V pipelined processors

Service / Outreach

Panini Linguistics Olympiad (PLO)

Problemsetter, 2025 & 2024

Student-Faculty Council, BITS Pilani – Hyderabad Campus

Acted as a mediator between departmental faculty and over 500 students for a selection of courses, to discuss ways to make the student experience better during the remote mode of study during the COVID-19 era

HSS F222 Introduction to Linguistics (Spring 2023)

CS F327 Computer Architecture (Spring 2022)

EEE F211 Electrical Machines (Spring 2021)

Rotaract Club, BITS Pilani – Hyderabad Campus, 2021

Hosted a professional development workshop for girls in STEM, with Chair, Grace Hopper Conference India

Languages

Programming: Python (Torch), R, MATLAB, C, C++, HTML/CSS, JavaScript, LaTeX

Natural Languages: English, Hindi-Urdu, Telugu, French (Functional), Latin (Basic)

Writing Systems: Latin, Devanagari, Telugu, Perso-Arabic, Brahmi

Industry Experience

Software Engineer

Centene Corporation

September 2024 – May 2025

St. Louis, MO

Data Scientist

Indeed, Inc.

May 2024 – July 2024

Stamford, CT

Software Engineering Intern

Dell EMC

July 2022 – December 2022

Bengaluru, India

Summer Intern

Edutech Learning Solutions

May 2021 – July 2021

Vadodara, India